

ECE 332 — Homework 4: Generated Answers from HW4 Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Symbol reference (used throughout):

- l — antenna physical length (m)
- $\lambda = c/f$ — wavelength (m); $c = 3 \times 10^8$ m/s
- I_0 — peak current at antenna feed (A)
- R — distance from antenna to observation point (m)
- θ — angle from antenna axis (rad or deg)
- $\eta_0 = 120\pi \approx 377 \Omega$ — intrinsic impedance of free space
- $k = 2\pi/\lambda$ — wavenumber (rad/m)
- $S(R, \theta)$ — average power density radiated (W/m²)
- $F(\theta) = S(\theta)/S_{\max}$ — normalized radiation intensity (unitless)
- β — half-power beamwidth (rad or deg)
- Ω_p — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$ — directivity (unitless, often dB)
- $A_e = \lambda^2 D / (4\pi)$ — effective aperture area (m²)
- A_p — physical cross-section area (m²)
- G_t, G_r — antenna gains (linear)
- P_t, P_r — transmitted / received power (W)

Key shortcuts: Hertzian dipole ($l/\lambda < 1/50$) $\Rightarrow D = 1.5$. Half-wave dipole ($l = \lambda/2$) $\Rightarrow D = 1.64$, $R_{\text{rad}} = 73 \Omega$. 3 dB $\Rightarrow$ linear gain of 2.

Problem 1 — Hertzian Dipole, Power Density at 5 km

8 pts

Setup: A 1-m long dipole excited by a 1 MHz current with amplitude $I_0 = 12$ A. Find the average power density radiated at $R = 5$ km, $\theta = 45^\circ$.

DIPOLE TYPE BY LENGTH

Check first *compute $\lambda = c/f$, then l/λ*

l/λ	Type	Use
$< 1/50$	Hertzian (short)	Hertzian formula
$= 1/4$	Quarter-wave	Arbitrary formula
$= 1/2$	Half-wave	$D = 1.64$, simple
other	Arbitrary	Arbitrary formula

If $l/\lambda < 1/50$ stop and use Hertzian only — don't subst. small l into arbitrary formula.

HERTZIAN DIPOLE — $S(R, \theta)$

Power density $l/\lambda < 1/50$; far field

$$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta \text{ (W/m}^2\text{)}$$

$k = 2\pi/\lambda$ rad/m; I_0 peak current (A); l length (m); R obs. dist (m).

Max direction $\theta = \pi/2$ (broadside)

Total radiated power *integrate over 4π sr*

$$P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda} \right)^2 \text{ (W)}$$

Directivity *Hertzian, exact*

$$D_{\text{Hertz}} = 1.5 \text{ (i.e. 1.76 dB)}$$

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{\text{rad}} (\Omega)$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

Step 1 — Classify the dipole length. Compute the wavelength to compare with l :

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^6} = 300 \text{ m}$$

$$\frac{l}{\lambda} = \frac{1}{300} < \frac{1}{50} \Rightarrow \text{Hertzian dipole}$$

The DIPOLE TYPE BY LENGTH section says short ($l/\lambda < 1/50$) means we can use the Hertzian $S(R, \theta)$ formula directly.

Step 2 — Apply the Hertzian power density formula. From HERTZIAN DIPOLE — $S(R, \theta)$:

$$S(R, \theta) = \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta$$

Step 3 — Plug in numbers. $\eta_0 = 120\pi$, $k = 2\pi/\lambda = 2\pi/300 \text{ rad/m}$, $I_0 = 12 \text{ A}$, $l = 1 \text{ m}$, $R = 5000 \text{ m}$, $\theta = \pi/4$:

$$S = \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5 \times 10^3)^2} \sin^2(\pi/4)$$

Numerator: $(120\pi)(4\pi^2/9 \times 10^4)(144) = (120\pi)(144)(4\pi^2)/(9 \times 10^4)$

$$= 120 \cdot 144 \cdot 4 \cdot \pi^3 / (9 \times 10^4) = 69120\pi^3 / (9 \times 10^4)$$

Denominator: $32\pi^2 \cdot 2.5 \times 10^7$. And $\sin^2(\pi/4) = 1/2$.

Working it out cleanly:

$$S(5 \text{ km}, 45^\circ) = 1.51 \times 10^{-9} \text{ W/m}^2$$

Sanity: a 12 A current at 1 MHz across a 1-m antenna at 5 km radiates 1.5 nW/m². The $1/R^2$ falloff dominates: most of the radiated power scatters into space, only a tiny fraction reaches any one observation point.

Problem 2 — Beam Parameters from Normalized Pattern

15 pts

Setup: $F(\theta) = \exp(-20\theta^2)$ for $0 \leq \theta \leq \pi$ (θ in radians). Find: (a) half-power beamwidth β , (b) pattern solid angle Ω_p , (c) directivity D .

BEAM PARAMETERS — HPBW, Ω_p , D

1. Normalize

$$F(\theta) = S(\theta)/S_{\max}$$

2. Half-power beamwidth (HPBW) *solve* $F(\theta_{1/2}) = 0.5$

$$\beta = 2\theta_{1/2} \quad (\text{symmetric beams})$$

example $F(\theta) = e^{-20\theta^2} \Rightarrow \theta_{1/2} = \sqrt{-\ln 0.5/20} = 0.186 \text{ rad} \Rightarrow \beta = 21.31^\circ$

3. Pattern solid angle (*sr*)

$$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$$

If F has no ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.

4. Directivity

$$D = \frac{4\pi}{\Omega_p} \quad (\text{unitless}) \quad D_{\text{dB}} = 10 \log_{10} D$$

5. Gain ξ = *radiation efficiency*, $0 < \xi \leq 1$

$$G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$$

For lossless / ideal antennas: $G = D$.

(a) **Half-power beamwidth β (5 pts).** HPBW is the angular width between the two points where $F(\theta) = 1/2$. Solve $\exp(-20\theta^2) = 0.5$:

$$-20\theta^2 = \ln(0.5) \Rightarrow \theta = \sqrt{-\ln(0.5)/20} = \sqrt{0.693/20} = \pm 0.186 \text{ rad}$$

The beam is symmetric about $\theta = 0$, so the full beamwidth is:

$$\boxed{\beta = 2\theta = 0.372 \text{ rad} = 21.31^\circ}$$

(b) **Pattern solid angle Ω_p (5 pts).** From BEAM PARAMETERS:

$$\Omega_p = \int \int_{4\pi} F(\theta) \sin \theta \, d\theta \, d\phi = \int_0^{2\pi} \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta \, d\phi$$

F has no ϕ dependence, so $\int_0^{2\pi} d\phi = 2\pi$:

$$\Omega_p = 2\pi \int_0^\pi \exp(-20\theta^2) \sin \theta \, d\theta$$

This integral has no closed form, evaluate numerically (Python/MATLAB/calculator):

$$\boxed{\Omega_p = 0.156 \text{ sr}}$$

(c) **Directivity D (5 pts).** From BEAM PARAMETERS:

$$D = \frac{4\pi}{\Omega_p} = \frac{4\pi}{0.156}$$

$$D = 80.55$$

Sanity: $D \approx 80$ is highly directive (much narrower than half-wave dipole's 1.64). The narrow Gaussian-like beam ($\beta \approx 21^\circ$) concentrates radiation into a small solid angle, so directivity is large.

Problem 3 — $\lambda/4$ Dipole Radiation Pattern

15 pts

Setup: Dipole of length $l = \lambda/4$. Find: (a) directions of maximum radiation, (b) expression for $S_{\max}$, (c) plot of normalized pattern $F(\theta)$.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at distance R any l

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Normalized pattern $F(\theta) = S(\theta)/S_{\max}$, dimensionless

Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta}, \quad D = 1.64, \quad R_{\text{rad}} = 73 \Omega$$

Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$$

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{\text{rad}} (\Omega)$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

General arbitrary-length dipole pattern. From ARBITRARY-LENGTH DIPOLE — $S(\theta)$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Plug in $l = \lambda/4$, so $\pi l/\lambda = \pi/4$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \cos(\pi/4)}{\sin \theta} \right]^2 = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Let $S_0 = 15I_0^2/(\pi R^2)$ be the constant prefactor.

(a) Direction of maximum radiation (5 pts). Maximizing $S(\theta)/S_0$ for $0 \leq \theta \leq \pi$. By calculus or numerical optimization, the max occurs at:

$$\theta_{\max} = \pi/2 = 90^\circ$$

(broadside to the dipole axis, same as half-wave). At $\theta = 0$ or π (along axis), $S = 0$.

(b) $S_{\max}$ (5 pts). Evaluate $S(\pi/2)$:

$$S_{\max} = S(\pi/2) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cdot 0\right) - \frac{\sqrt{2}}{2}}{\sin(\pi/2)} \right]^2 = \frac{15I_0^2}{\pi R^2} \left(1 - \frac{\sqrt{2}}{2}\right)^2$$

Numerically $(1 - \sqrt{2}/2)^2 = 0.0858$:

$$S_{\max} = 0.0858 \frac{15I_0^2}{\pi R^2}$$

(c) **Normalized pattern** $F(\theta) = S(\theta)/S_{\max}$ (5 pts):

$$F(\theta) = \frac{1}{0.0858} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Bell-shaped curve peaking at $\theta = \pi/2$ with $F = 1$, dropping to 0 at $\theta = 0$ and π . Symmetric about $\theta = \pi/2$. Similar shape to half-wave dipole pattern but slightly narrower since the dipole is shorter.

Sanity: A $\lambda/4$ dipole has a broadside pattern very close to a Hertzian dipole's $\sin^2 \theta$ shape. As $l \rightarrow 0$ the bracket approaches $\sin \theta$ and $S \propto \sin^2 \theta$ (Hertzian). Half-wave $l = \lambda/2$ has slightly narrower beam than the $\lambda/4$ case.

Problem 4 — Effective Area vs. Physical Cross-section

5 pts

Setup: Half-wave dipole at $f = 100$ MHz, wire diameter $d = 2$ cm. Find effective area A_e and compare to physical cross-section A_p .

EFFECTIVE AREA A_e	COMMON DIPOLE PARAMETERS																				
Definition m^2, antenna's receiving cross section $A_e = \frac{\lambda^2 D}{4\pi}$ Physical cross section half-wave dipole, wire diameter d $A_p = l\,d = \frac{\lambda}{2}\,d$ Ratio $A_e \gg A_p$ for thin wires (good!) Example: $f = 100\text{ MHz}$, $d = 2\text{ cm} \Rightarrow A_e/A_p \approx 39$.	<table><tr><th>Type</th><th>D</th><th>D_{dB}</th><th>$R_{\text{rad}}\ (\Omega)$</th></tr><tr><td>Isotropic</td><td>1</td><td>0</td><td>—</td></tr><tr><td>Hertzian</td><td>1.5</td><td>1.76</td><td>$80\pi^2(l/\lambda)^2$</td></tr><tr><td>Half-wave</td><td>1.64</td><td>2.15</td><td>73</td></tr><tr><td>$l = \lambda$</td><td>2.41</td><td>3.82</td><td>199</td></tr></table>	Type	D	D_{dB}	$R_{\text{rad}}\ (\Omega)$	Isotropic	1	0	—	Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$	Half-wave	1.64	2.15	73	$l = \lambda$	2.41	3.82	199
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Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3 \text{ m}$$

Step 2 — Directivity for half-wave dipole. From COMMON DIPOLE PARAMETERS: $D = 1.64$.

Step 3 — Effective area. From EFFECTIVE AREA A_e :

$$A_e = \frac{\lambda^2 D}{4\pi} = \frac{(3)^2 (1.64)}{4\pi} = \frac{14.76}{4\pi}$$

$$A_e = 1.17 \text{ m}^2$$

Step 4 — Physical cross-section. A half-wave dipole has length $l = \lambda/2$. Treat as a rectangle of length l and width d :

$$A_p = l \cdot d = \frac{\lambda}{2} \cdot d = \frac{3}{2} (0.02) = 0.03 \text{ m}^2$$

Step 5 — Ratio.

$$\frac{A_e}{A_p} = \frac{1.17}{0.03} = 39.2$$

Sanity: The effective area is $39\times$ larger than the physical cross-section. This is the key insight of antenna theory: a thin wire antenna captures energy from a much larger "effective" patch of the incoming wave than its physical wire would suggest, because the antenna interacts with the field over a region $\sim \lambda$ around itself.

Problem 5 — Friis Transmission, TV Broadcast

5 pts

Setup: Half-wave dipole TX, 1 kW at 50 MHz. RX home antenna with 3 dB gain, distance 30 km. Find received power.

FRIIS TRANSMISSION FORMULA

Far-field link budget P_t tx, P_r rx, dist R

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

G_t, G_r linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.
Solve for R

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$$

Solve for G_r given A_e

$$G_r = \frac{4\pi A_{e,r}}{\lambda^2} \Rightarrow P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$$

Use linear G , not dB. Convert first: $G_{lin} = 10^{G_{dB}/10}$.Free-space path loss $L_{fs} = (4\pi R/\lambda)^2$ so $P_r = G_t G_r P_t / L_{fs}$.

dB ↔ LINEAR

Power gain factor → dB

$$G_{dB} = 10 \log_{10} G_{lin} \quad G_{lin} = 10^{G_{dB}/10}$$

	dB	linear	dB	linear
	0	1	10	10
	3	2	13	20
	6	4	20	100
	7	5	30	1000
	-3	0.5	-10	0.1

For E or amplitude use $20 \log$ not $10 \log$.

Step 1 — Wavelength.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{5 \times 10^7} = 6 \text{ m}$$

Step 2 — Antenna gains. TX is half-wave dipole, so $G_t = 1.64$ (from COMMON DIPOLE PARAMETERS).

RX gain: 3 dB → linear via $G_{lin} = 10^{G_{dB}/10}$:

$$G_r = 10^{3/10} = 2$$

(3 dB always = $\times 2$, this is the cleanest dB conversion to remember.)

Step 3 — Friis formula. From FRIIS TRANSMISSION FORMULA:

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

Step 4 — Plug in. $R = 30 \text{ km} = 3 \times 10^4 \text{ m}$, $P_t = 10^3 \text{ W}$:

$$P_r = (1.64)(2) \left(\frac{6}{4\pi(3 \times 10^4)} \right)^2 (10^3)$$

The path-loss factor:

$$\frac{\lambda}{4\pi R} = \frac{6}{4\pi \cdot 3 \times 10^4} = \frac{6}{3.77 \times 10^5} = 1.59 \times 10^{-5}$$

$$\left(\frac{\lambda}{4\pi R} \right)^2 = 2.53 \times 10^{-10}$$

$$P_r = (1.64)(2)(2.53 \times 10^{-10})(10^3) = (3.28)(2.53 \times 10^{-7})$$

$$P_r = 8.3 \times 10^{-7} \text{ W} = 0.83 \mu\text{W}$$

Sanity: 1 kW transmitted, 0.83 μW received at 30 km. That's a 90 dB drop, dominated by the $1/R^2$ free-space path loss. Plenty of signal for a TV receiver (typical sensitivity is $-100 \text{ dBm} = 10^{-13} \text{ W}$), so this link closes comfortably.

Note on the solution PDF: the official key wrote “(1.64)(3)” as a typo, but used $G_r = 2$ in the actual arithmetic (since 3 dB → linear = 2). The final answer $8.3 \times 10^{-7} \text{ W}$ matches either way.